

Date	15 March 2026
Team ID	XXXXXX
Project Name	VBCUA — Voice-Based Concept Understanding Analyser
Maximum Marks	4 Marks

SOLUTION REQUIREMENTS

DEFINE FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).

Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

STEP 1: FUNCTIONAL REQUIREMENTS (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	The system operates without complex user authentication in its initial local desktop configuration; users immediately access the primary Streamlit UI application upon launch.	Low
2	Authorization levels	Features a flat single-user role architecture by default, granting immediate access to both the assessment execution tab and the comprehensive history dashboard.	Low
3	External interfaces	Connects locally to OpenAI Whisper for speech transcription, Sentence-BERT for similarity evaluations, and securely hooks into the external Google Gemini API for AI feedback.	High
4	Transactions processing	Accepts spoken explanation audio file uploads (WAV, MP3, M4A) via Streamlit, routes them through the ML pipeline, and writes metadata records into a local SQLite data store.	High
5	Reporting	Displays instant metrics grids (Semantic, Fluency, Overall, Grade, and WPM details), plots audio waveform charts, and compiles downloadable PDF analytical review sheets.	High
6	Business rules, etc.	Generates final assessment values dynamically using a weighted composite ratio logic (60% Semantic weight and	High

S.No	Requirement Category	Requirement Description	Priority
		40% Fluency weight) against an ideal 110–160 WPM delivery speed.	
7	Compliance to laws or regulations	Mandates localized data containment workflows where local database files, logs, and transient media assets are protected via strict runtime <code>.env</code> structures and automated <code>.gitignore</code> policies.	Medium
8	Other	Employs an automated rule-based analytics engine fallback that engages seamlessly if the primary Google Gemini API connection is completely absent or fails.	Medium

STEP 2: NON-FUNCTIONAL REQUIREMENTS (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Optimizes localized ML evaluation runtimes and structures prompt interface updating frequencies.	Whisper transcription and Sentence-BERT vector matching complete within 10 seconds per minute of uploaded media.
2	Scalability	Architects data structures to scale smoothly from isolated local desktop sessions into heavy multi-user enterprise frameworks.	Supports completely transparent migrations to external Postgres servers by adjusting a single runtime environment variable.
3	Security & Data Privacy	Shields infrastructure endpoints and completely separates external API access credentials from public application repositories.	Application environment variables and authorization strings are entirely managed via detached, uncommitted local environmental files.
4	Reliability & Availability	Safeguards calculation pipelines and guarantees continuous operational analysis capacities during peripheral cloud disconnects.	The alternative internal feedback handler keeps processing pipelines active at a 100% operational rate during API network timeouts.
5	Usability & Accessibility	Delivers a clean, highly scannable, component-driven dashboard design layout divided into explicit "Analyzer" and "Dashboard" tabs.	Streamlit responsive application framework yields full compatibility across various device orientations and distinct viewport resolutions.
6	Other	Mandates clear separation of execution concerns to ensure long-term codebase maintainability and simple functional testing.	Modular directory organization separating user interfaces, core algorithms, and reports (src /, data /, reports /) with test suite integration.